MTIA: First Generation Silicon Targeting Meta's Recommendation Systems

Sparsh Mittal

Based on a paper published in ISCA 2023

Workloads

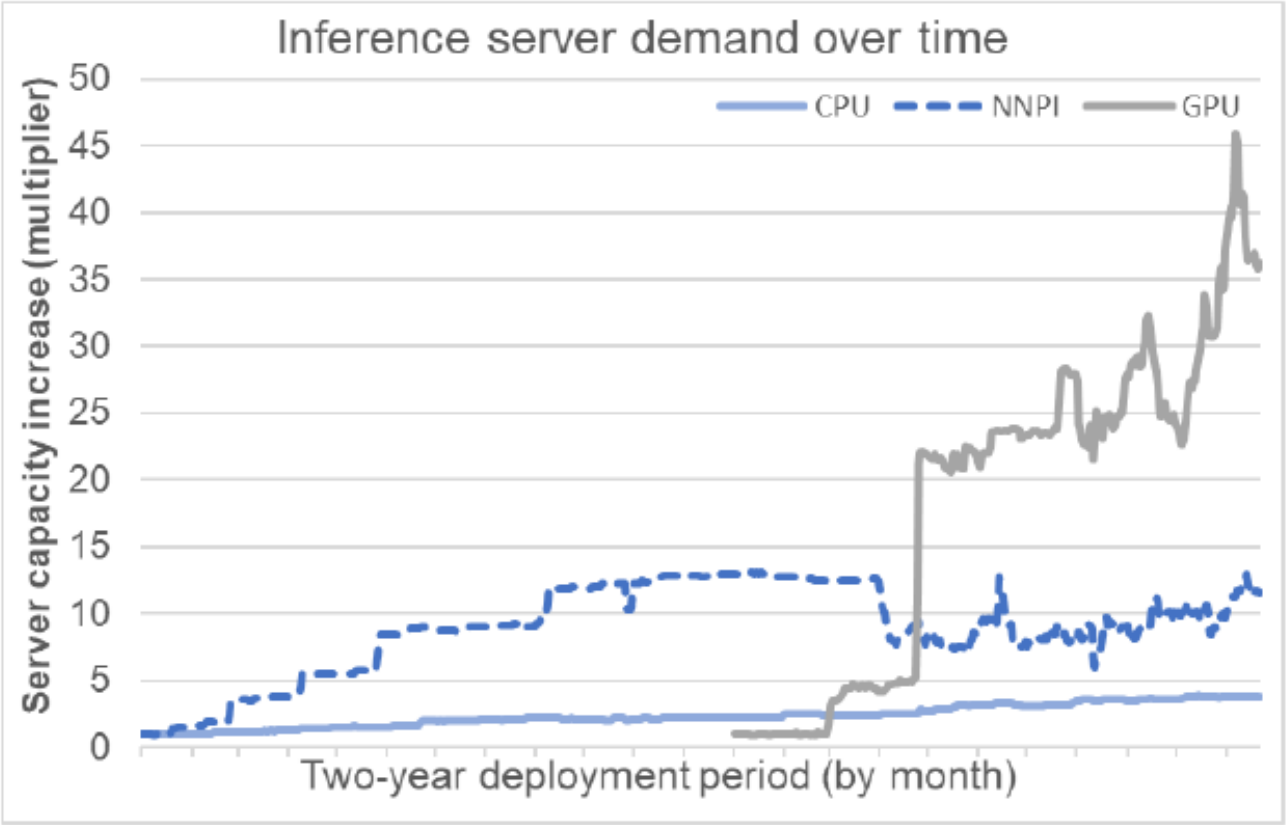
Deep Learning Recommendation Models (DLRM) are the most dominant workloads in Meta's datacenters.

DLRM combine MLP (which are compute-intensive) with embedding tables that transform sparse features into a dense representation. Embedding table operations are memory-intensive.

Motivation

Traditionally, CPUs were used, but they are not cost-effective.

- Dashed line: Servers equipped with the first-generation inference accelerator Intel NNPI
- Dark solid line: GPU-based servers
- The initial demand for increased capacity was temporarily met using the NNPI accelerator, the requirements for the inference models quickly outpaced the NNPI capabilities
- The increased demand was met using GPUs



Growth in server demand for inference workloads

Recent generations of GPUs provide a lot of memory bandwidth and compute power

However, they are not designed with inference in mind, and therefore the efficiency of processing real inference workloads is low.

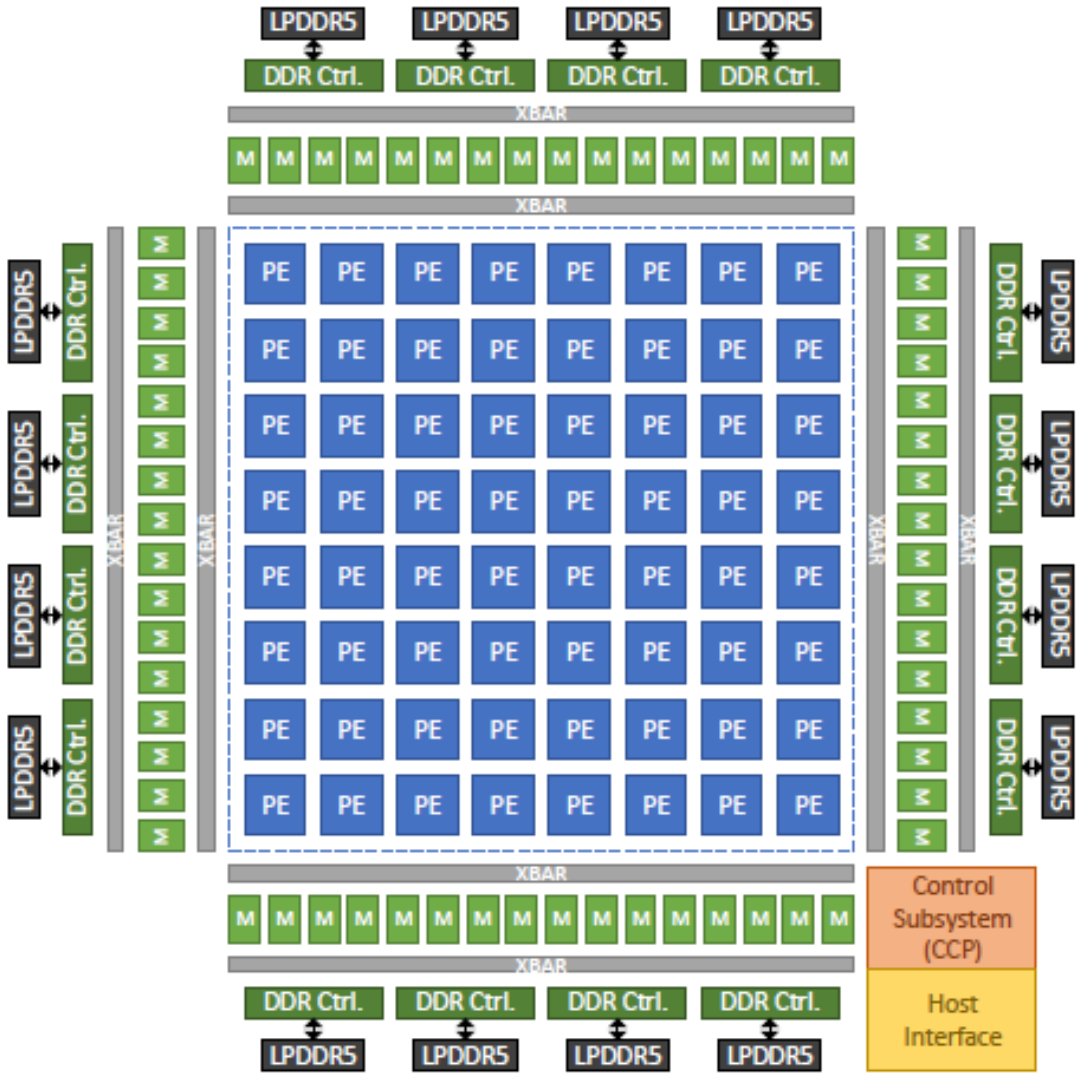
High-level architecture

It is an array of processing elements (PEs) connected on a grid.

The grid is connected to a set of on-chip memory blocks and off-chip memory controllers through crossbars on each side.

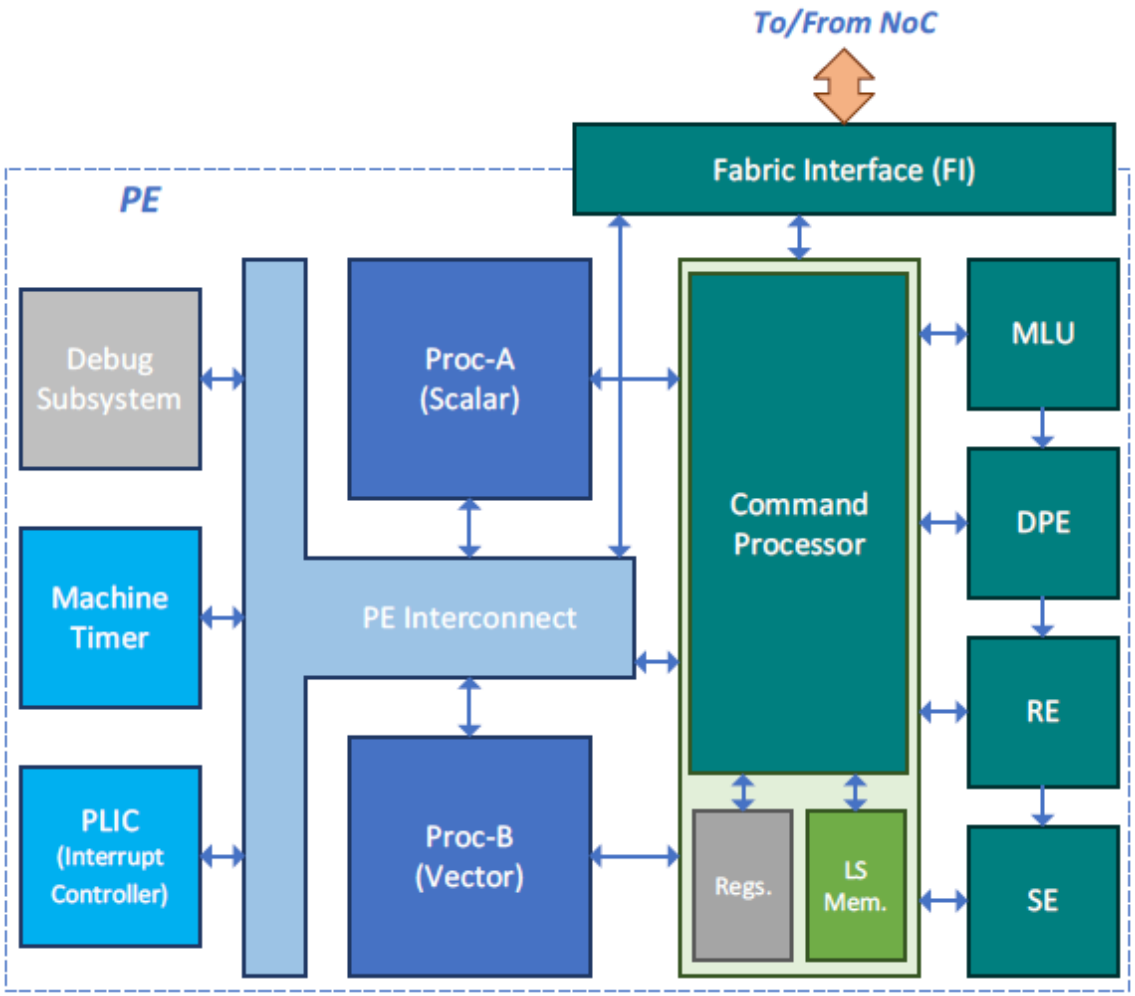
There is a separate control subsystem with dedicated processors and peripherals to run the system's control software.

The host interface unit contains a PCIe interface



Architecture of each PE

- A PE has
  - two RISC-V processor cores and associated peripherals (on the left),
  - 5 fixed function units specialized in performing specific computations or data movements (on the right).
  - 128KB of local storage.
  - Command Processor which orchestrates and coordinates execution of operations on these fixed function blocks



Fixed-function blocks

A **local interconnect** establishes the connectivity between processors, their peripherals and custom hardware blocks.

Memory-layout unit:

operates on tensors with 4/8/16/32-bit data types

Performs operations like copying, transpose, concatenation, or reshape

Dot-Product Engine (DPE) performs 1024 INT8 multiplications (32×32) or 512 FP16/BF16 multiplications (32×16) per cycle.

Reduction engine: The reduction engine hosts the storage elements that keep track of the results of the matrix multiplication operations and accumulates them over multiple operations.

SIMD Engine (SE)

This block performs operations like quantization/dequantization and nonlinear functions.

It has lookup tables and floating-point arithmetic units to calculate linear or cubic approximation of nonlinear functions such as exponentials, sigmoid, tanh, etc.

The approximation accepts INT8 or FP16 data types as inputs, producing an INT8 or FP32 result at the output.

It can also use its floating-point ALUs to perform predefined elementwise operations, such as addition, multiplication, accumulation, etc.

Fabric Interface (FI)

It is the gateway in and out of the PE.

It connects to and communicates over the accelerator's on-chip network.

It sends memory access requests to on-chip and offchip memories, and system registers, and receives back the data or write completions.

Processor cores

- Each PE has two RISC-V cores that run the application's code and issue commands to the CP for offloading various computations to fixed function units.
- The cores are single issue, in-order cores, with a five-stage pipeline.
- They also have custom interfaces, custom registers, custom instructions, etc.
  - For example, custom instructions are added to start the desired operation on each of the fixed function units.
- One of the processor cores supports RISC-V vector extension, which adds extra flexibility to the PE and allows implementing operations that do not map well to the existing fixed function units.
- The vector processing unit contains 32 vector registers, each 64B wide.

Local memories can be accessed by cores via regular load/store instructions.

The MATMUL instruction is stalled by the Command Processor until enough data is brought into the local memory by prior DMA (direct memory access) operations, and is started immediately afterwards, relieving the program from explicitly checking the availability of the data.

Memory subsystem

MTIA also has 128MB of on-chip SRAM, organized as slices around the grid.

This on-chip memory can be used as addressable scratchpad memory, or as a common, shared, memory-side cache.

There are four LPDDR5 controllers on each side of the grid, providing a peak 176 GB/s off-chip bandwidth.

Off-chip memory can be upto 128GB.

Memory addresses are distributed across these controllers, and among the on-chip SRAM slices.

An AXI-based on-chip network connects all the PEs and memories together.

Caching optimization

Processor cores have instruction and data caches

There is a large on-chip last level cache

Input operands of DPE are cached

PEs can store operands in local memories.

Parallelism leveraged at different levels:

Data-parallelism:

Wide vectors in fixed function units

Use of vector processors.

Multiple PEs can operate on the same task in a data parallel manner.

Instruction level parallelism:

The Command Processor allows multiple outstanding operations to be handled by different fixed function blocks simultaneously.

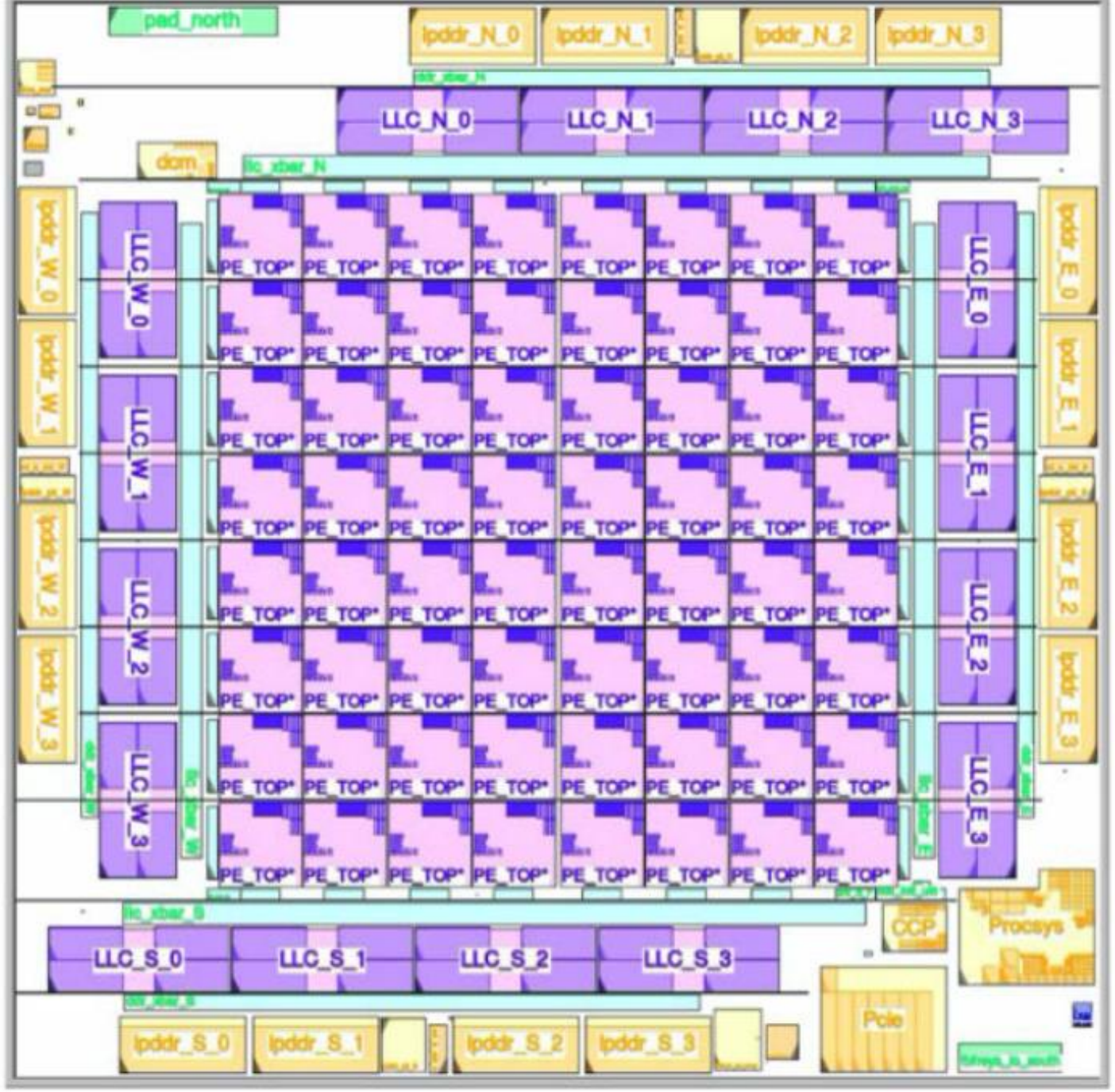
Memory level parallelism:

many outstanding requests can go to on-chip and off-chip memories from each PE.

Thread level parallelism:

Each PE runs two threads: one for data-movement and one for computation.

Die plot


pad_north
lpddr_N_0
lpddr_N_1
lpddr_N_2
lpddr_N_3
LLC_N_0
LLC_N_1
LLC_N_2
LLC_N_3
dom
llc_xbar_N
lpddr_W_0
lpddr_W_1
lpddr_W_2
lpddr_W_3
LLC_W_0
LLC_W_1
LLC_W_2
LLC_W_3
PE_TOP*
LLC_E_0
LLC_E_1
LLC_E_2
LLC_E_3
lpddr_E_0
lpddr_E_1
lpddr_E_2
lpddr_E_3
llc_xbar_S
LLC_S_0
LLC_S_1
LLC_S_2
LLC_S_3
CCP
Procsys
Pcie
lpddr_S_0
lpddr_S_1
lpddr_S_2
lpddr_S_3

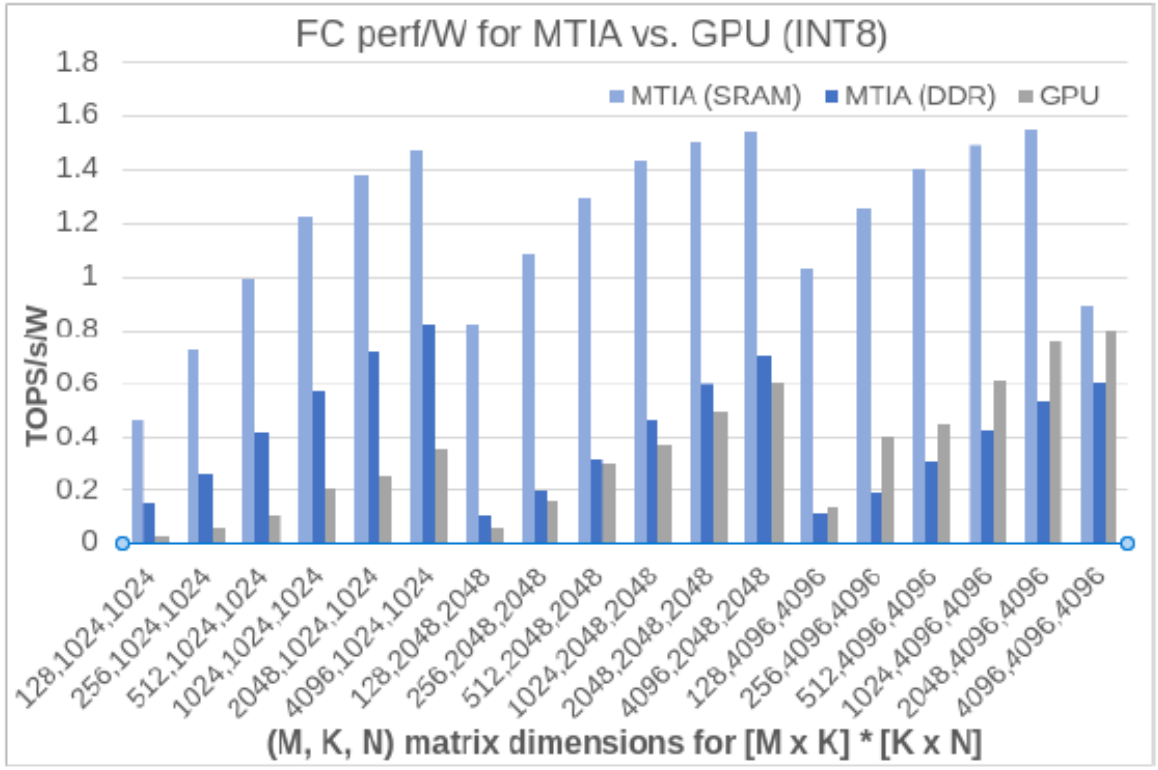
Parameter	Value
Technology	TSMC 7nm
Frequency	800MHz nominal (1.1 GHz max)
Instances	1.12B gates, 65M flops
Dimensions	19.34 × 19.1mm (373 mm^2)
Package	43 × 43, ~2800 pins
TDP	25 W
Voltage	Dual rail: 0.67V (logic), 0.75V (memories)
Host Connectivity	8× PCIe Gen4 (16 GB/s)
GEMM TOPS (MAC)	102.4 (INT8)
	51.2 (FP16)
SIMD TOPS	Vector: 0.8 (FP32) / 1.6 (FP16) / 3.2 (INT8)
	SE: 1.6 (FP16) / 3.2 (INT8)
Memory Bandwidth	Local memory: 400GB/s per PE
	On-chip SRAM: 800GB/s
	Off-chip DRAM: 176 GB/s
Memory Capacity	Local memory: 128KB per PE
	On-chip SRAM: 128MB
	Off-chip LPDDR5: 64GB (16 channels)

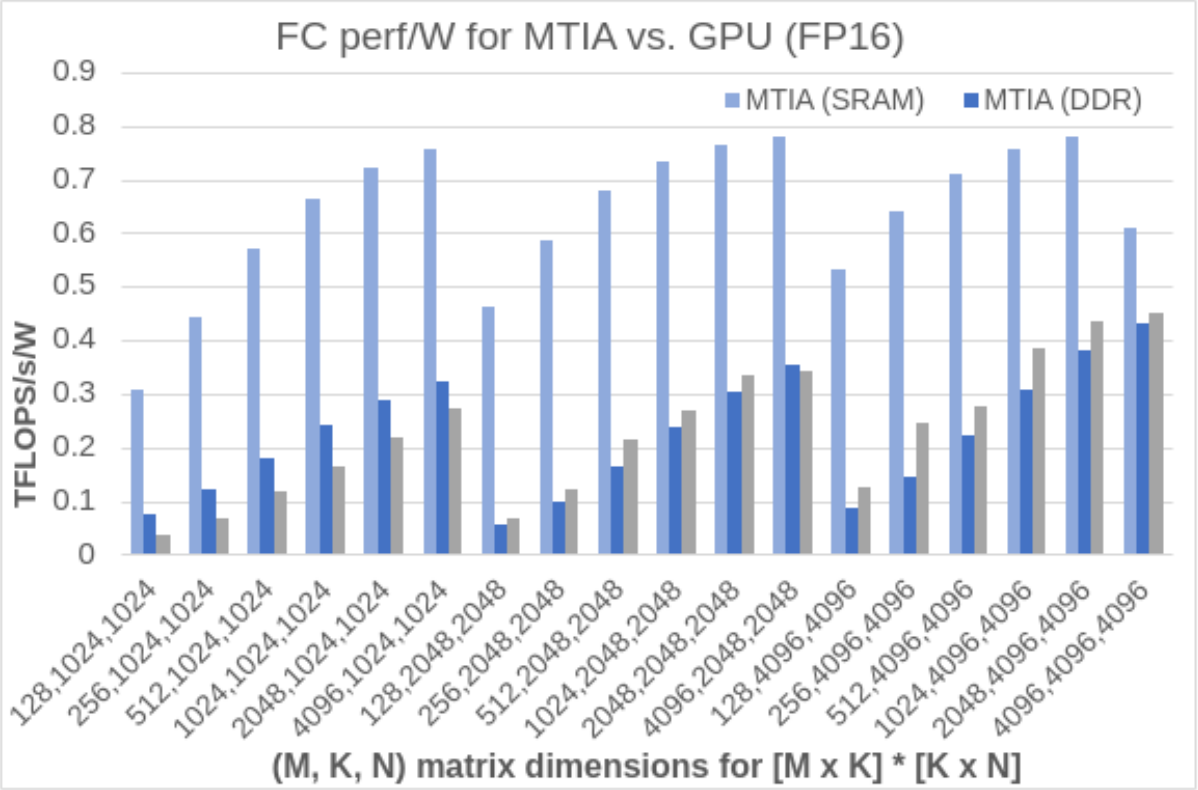
Inference hardware platforms

Metric		Yosemite V2 (6 NNPI)	Zion4S (8 GPU)	Yosemite V3 (12 MTIA)
Power	System	298 W	4500 W	780 W
	Card	13.5 W	330 W	35 W
	Percentage	27.2 %	58.7 %	53.8 %
Compute	INT8 (TOPS/s)	50 × 6	624 × 8	104 × 12
	FP16 (TF/s)	6.25 × 6	312 × 8	52 × 12
Memory	Type (device)	LPDDR	HBM	LPDDR
	Size (device)	16 GB × 6	40 GB × 8	32 GB × 12
	BW (device)	50 GB/s × 6	1.5 TB/s × 8	150 GB/s × 12
	Size (host)	64 GB	1.5 TB	96 GB
	BW (host)	50 GB/s	400 GB/s	76 GB/s
Comms.	Dev.-to-Dev.	PCIe	NVLink	PCIe
	P2P BW (card)	3.2 GB/s	80 GB/s	12.8 GB/s
	NIC BW	50 Gbps	400 Gbps	100 Gbps

FC=fully-connected

- Int8 gives 2X performance than FP16
- In many cases, MTIA performance/watt >2X of GPU
- MTIA is particularly effective for low batch sizes.
- For large batch sizes, GPU achieves higher utilization, so MTIA has lower gains.
- MTIA is most efficient when tensors are accessed from SRAM.
- ==> graph optimizations and managing data locality are very important



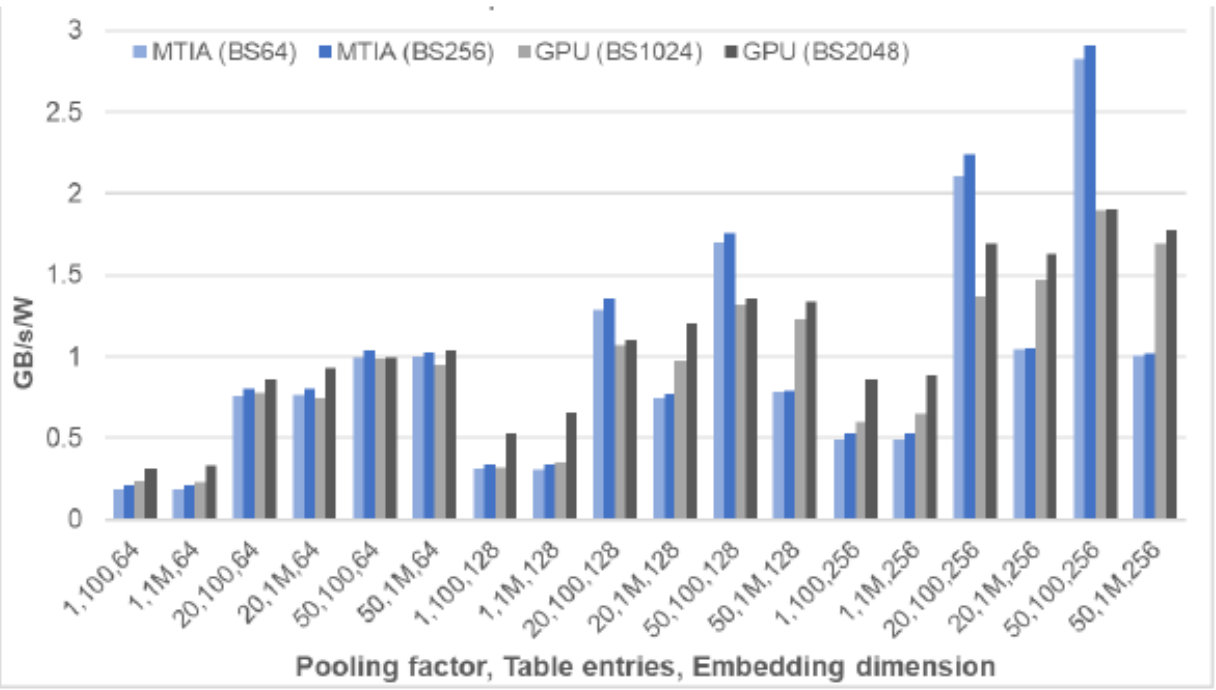


Sparse computation results

MTIA achieves between 0.6x to 1.5x the perf/W of the GPU with the current kernel implementation.

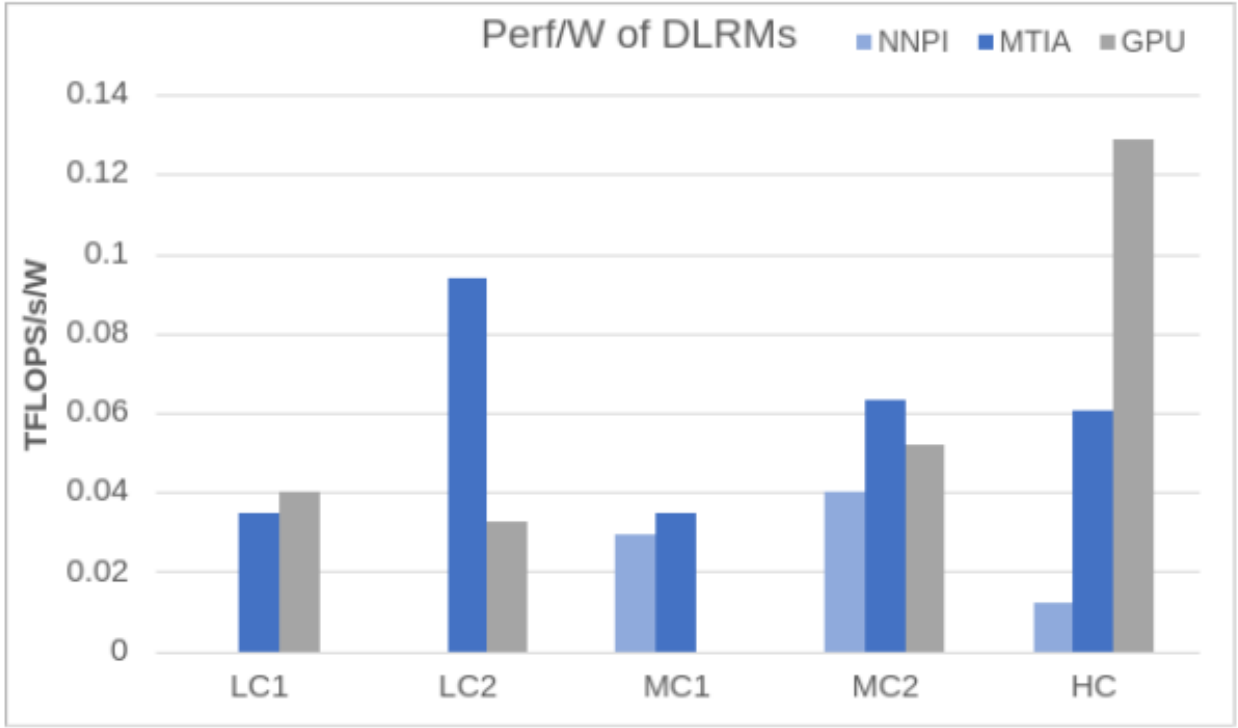
Software stack of MTIA is not mature.

MTIA is reaching just 10-20% of its memory bandwidth whereas the GPU is achieving about 60% of its HBM bandwidth.



MTIA: 1.6X than NNPI and 0.9x than GPU.

MTIA is good for LC models since they are dominated by FC layers with smaller input shapes.

For HC models, GPU software stack is better optimized for large shapes.

DLRM Model	Size (GB)	Complexity (GFLOPS/batch)
Low Complexity 1 (LC1)	53.2	0.032
Low Complexity 2 (LC2)	4.5	0.014
Medium Complexity 1 (MC1)	120	0.140
Medium Complexity 2 (MC2)	200	0.220
High Complexity (HC)	725	0.450